(iv) $\frac{1}{f(x)}$ is also continuous at $x = a$, if $f(a) \neq 0$.

(v) $\frac{f(x)}{g(x)}$ is also continuous at $x = a$, if $g(a) \neq 0$.

(vi) $|f(x)|$ is also continuous at $x = a$.

Example 5.27

Show that

$$f(x) = \begin{cases} 5x - 4, & if\ 0 < x \leq 1 \\ 4x^3 - 3x, & if\ 1 < x < 2 \end{cases}$$ is continuous at $x = 1$

Solution

$$L[f(x)]_{x=1} = \lim_{x \to 1^-} f(x)$$

$$= \lim_{h \to 0} f(1-h),\ x = 1-h$$

$$= \lim_{h \to 0} [5(1-h) - 4]$$

$$= 5(1) - 4 = 1$$

$$R[f(x)]_{x=1} = \lim_{x \to 1^+} f(x)$$

$$= \lim_{h \to 0} f(1+h),\ x = 1+h$$

$$= \lim_{h \to 0} [4(1+h)^3 - 3(1+h)]$$

$$= 4(1)^3 - 3(1)$$

$$= 4 - 3 = 1$$

Now, $f(1) = 5(1) - 4 = 5 - 4 = 1$

$\therefore \lim_{x \to 1^-} f(x) = \lim_{x \to 1^+} f(x) = f(1)$,

$f(x)$ is continuous at $x = 1$

Example 5.28

Verify the continuity of the function $f(x)$ given by

$$f(x) = \begin{cases} 2 - x\ if\ x < 2 \\ 2 + x\ if\ x \geq 2 \end{cases}$$ at $x = 2$

Solution

$$L[f(x)]_{x=2} = \lim_{x \to 2^-} f(x)$$

$$= \lim_{h \to 0} f(2-h),\ x = 2 - h$$

$$= \lim_{h \to 0} \{2 - (2-h)\}$$

$$= \lim_{h \to 0} h = 0$$

$$R[f(x)]_{x=2} = \lim_{x \to 2^+} f(x)$$

$$= \lim_{h \to 0} f(2+h),\ x = 2 + h$$

$$= \lim_{h \to 0} \{2 + (2+h)\}$$

$$= \lim_{h \to 0} (4 + h) = 4$$

Now, $f(2) = 2 + 2 = 4$

Here $\lim_{x \to 2^-} f(x) \neq \lim_{x \to 2^+} f(x)$

Hence $f(x)$ is not continuous at $x = 2$

Observations

(i) Constant function is everywhere continuous.

(ii) Identity function is everywhere continuous.

(iii) Polynomial function is everywhere continuous.

(iv) Modulus function is everywhere continuous.

(v) Exponential function a^x, $a>0$ is everywhere continuous.

(vi) Logarithmic function is continuous in its domain.

(vii) Rational function is continuous at every point in its domain.

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